

Bangladesh University of Business & Technology (BUBT)



Lab Report

Course Title : Md. Takwat Hossain

Course Code : CSE102

Lab Report No : 01

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Question 01 :Write a C program to find maximum between two numbers.

```
#include<stdio.h>

int main(){

int x,y;

printf("Enter two numbers: ");

scanf("%d %d",&x,&y);

if(x>y){

    printf("The maximum number: %d",x);

}

else{

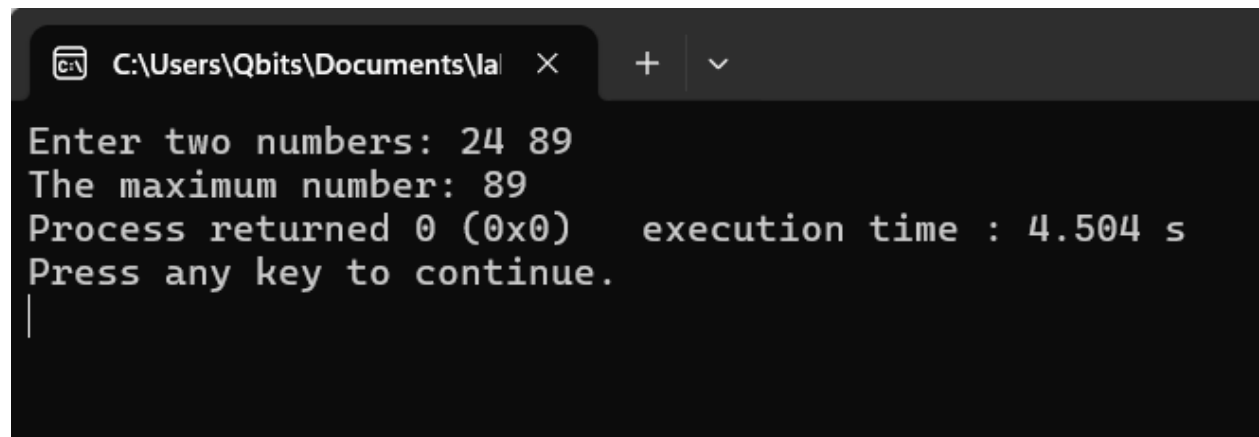
    printf("The maximum number: %d",y);

}

return 0;

}
```

Output:



```
C:\Users\Qbits\Documents\la X + v
Enter two numbers: 24 89
The maximum number: 89
Process returned 0 (0x0) execution time : 4.504 s
Press any key to continue.
|
```

Question 2: Write a C program to find maximum between three numbers.

```
#include<stdio.h>

int main(){

int x,y,z;

printf("Enter three numbers: ");

scanf("%d%d%d",&x,&y,&z);

if(x>=y && x>=z){

    printf("The maximum number %d",x);

}

else if(y>=x && y>=z){

    printf("The maximum number %d",y);

}

else{

    printf("The maximum number is %d",z);

}

return 0;

}
```

Output:

```
C:\Users\Qbits\Documents\la | X + v
Enter three numbers: 34 67 98
The maximum number is 98
Process returned 0 (0x0)    execution time : 5.311 s
Press any key to continue.
|
```

Question 3: Write a C program to check whether a number is negative, positive or zero.

```
#include<stdio.h>

int main(){

int x;

printf("Give a number: ");

scanf("%d",&x);

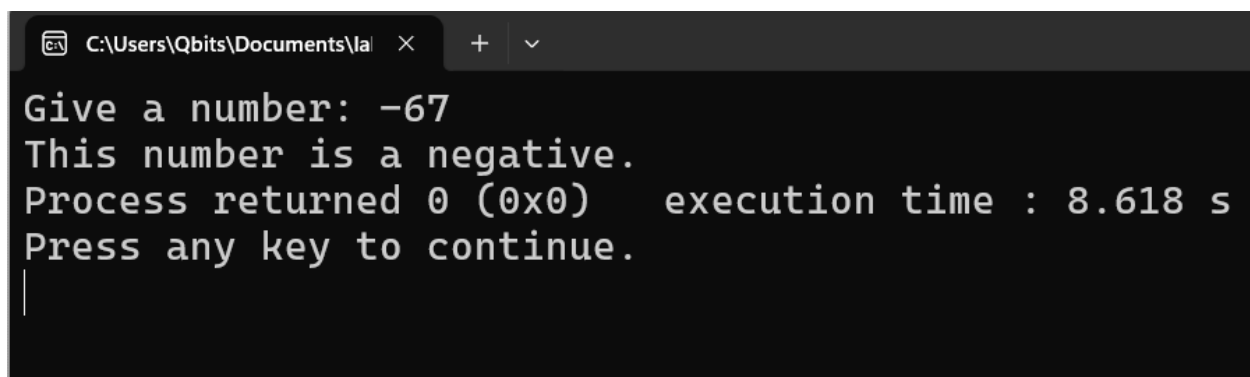
if(x>0){

    printf("This number is positive.");

}
```

```
else if(x<0){  
    printf("This number is a negative.");  
}  
else{  
    printf("This number is zero.");  
}  
return 0;  
}
```

Output:

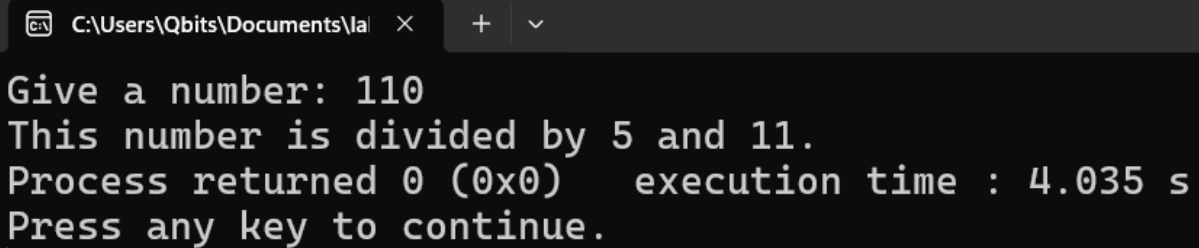
A screenshot of a terminal window with a dark background. The window title bar shows the file path 'C:\Users\Qbits\Documents\la' and standard window controls. The terminal output is as follows:
Give a number: -67
This number is a negative.
Process returned 0 (0x0) execution time : 8.618 s
Press any key to continue.
|

Question 4: Write a C program to check whether a number is divisible by 5 and 11 or not.

```
#include<stdio.h>
```

```
int main(){  
    int x;  
    printf("Give a number: ");  
    scanf("%d",&x);  
    if(x%5==0 && x%11==0){  
        printf("This number is divided by 5 and 11.");  
    }  
    else{  
        printf("This number is not devided by 5 and 11.");  
    }  
    return 0;  
}
```

Output :



The screenshot shows a Windows command prompt window with a dark background. The title bar at the top indicates the file path 'C:\Users\Qbits\Documents\la'. The command prompt displays the output of the program: 'Give a number: 110', 'This number is divided by 5 and 11.', 'Process returned 0 (0x0) execution time : 4.035 s', and 'Press any key to continue.' followed by a vertical cursor line.

```
Give a number: 110  
This number is divided by 5 and 11.  
Process returned 0 (0x0) execution time : 4.035 s  
Press any key to continue.  
|
```

Question 5: Write a C program to check whether a number is even or odd .

```
#include<stdio.h>

int main(){

int x;

printf("Give a number: ");

scanf("%d",&x);

if(x%2==0){

    printf("This number is an even number.");

}

else{

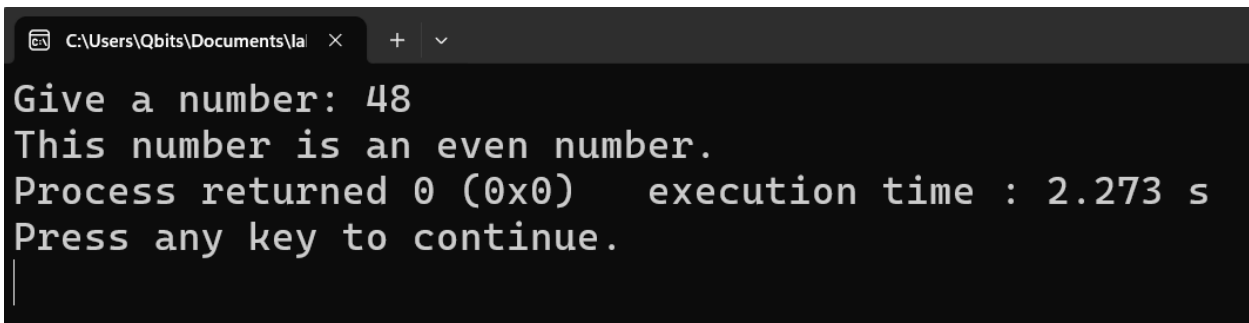
    printf("This number is an odd number.");

}

return 0;

}
```

Output :



```
C:\Users\Qbits\Documents\la  × + ▾
Give a number: 48
This number is an even number.
Process returned 0 (0x0)   execution time : 2.273 s
Press any key to continue.
|
```

Question 6: Write a C program to check whether a year is leap year or not.

```
#include<stdio.h>

int main(){

int x;

printf("Enter a year: ");

scanf("%d",&x);

if((x%400==0) || (x%4==0 && x%100!=0)){

    printf("This year is a leap year.");

}

else{

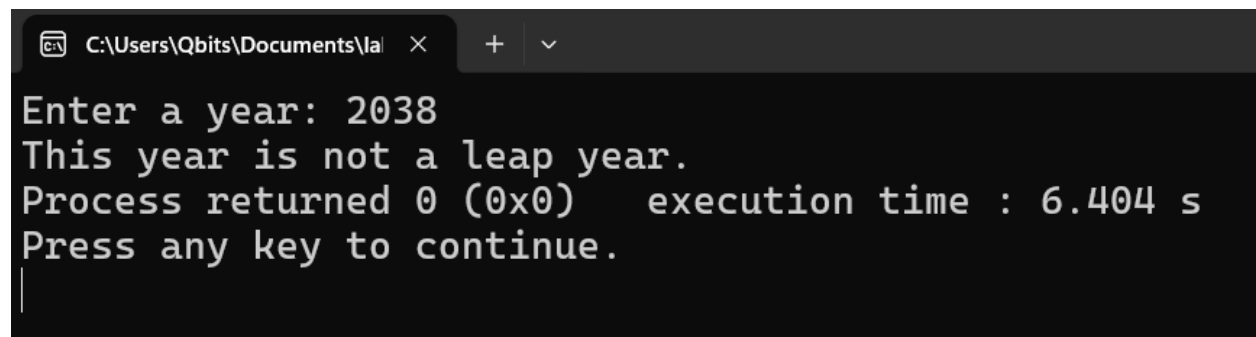
    printf("This year is not a leap year.");

}

return 0;

}
```

Output:



```
C:\Users\Qbits\Documents\la | X + v
Enter a year: 2038
This year is not a leap year.
Process returned 0 (0x0)    execution time : 6.404 s
Press any key to continue.
|
```


Question 7: Write a C program to check whether a character is alphabet or not.

```
#include<stdio.h>

int main(){

char x;

printf("Give a character: ");

scanf("%c",&x);

if((x>='a' && x<='z') || (x>='A' && x<='Z')){

    printf("This is an alphabet.");

}

else{

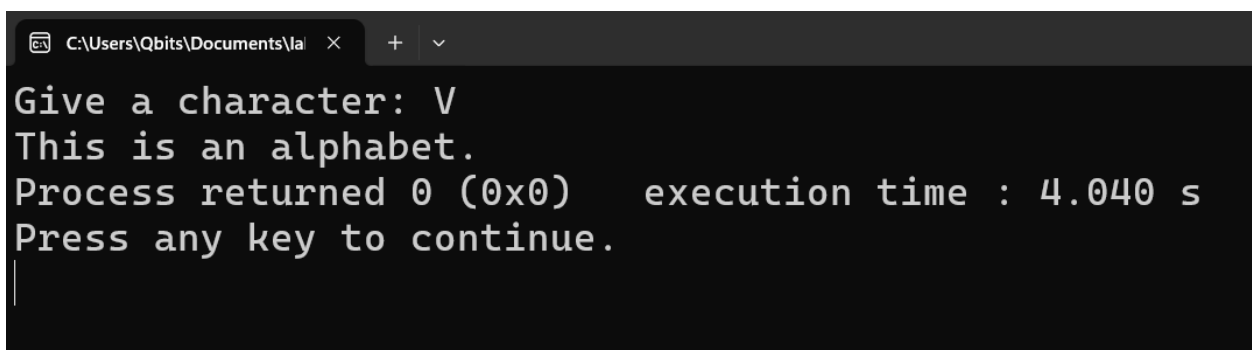
    printf("This is not an alphabet.");

}


return 0;

}
```

Output:



```
C:\Users\Qbits\Documents\la  ×  +  v
Give a character: V
This is an alphabet.
Process returned 0 (0x0)   execution time : 4.040 s
Press any key to continue.
|
```

Question 8: Write a C program to input any alphabet and check whether it is vowel or consonant.

```
#include <stdio.h>
```

```
int main() {
```

```
    char ch;
```

```
    printf("Enter any alphabet: ");
```

```
    scanf("%c", &ch);
```

```
    ch = tolower(ch);
```

```
    if (ch >= 'a' && ch <= 'z') {
```

```
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
```

```
            printf("%c is a vowel.\n", ch);
```

```
        } else {
```

```
            printf("%c is a consonant.\n", ch);
```

```
        }
```

```
    } else {
```

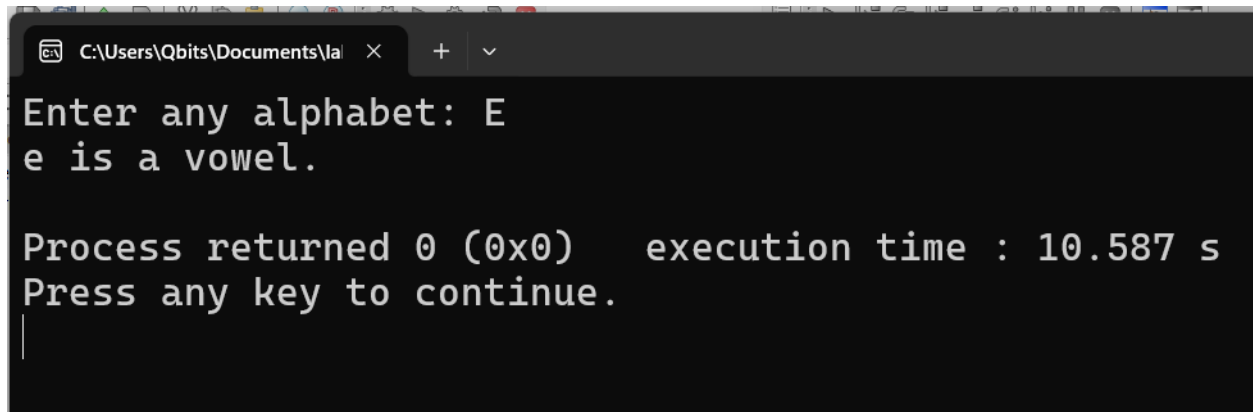
```
        printf("Invalid input! Please enter an alphabet.\n");
```

```
    }
```

```
    return 0;
```

```
}
```

Output:



```
C:\Users\Qbits\Documents\la
Enter any alphabet: E
e is a vowel.

Process returned 0 (0x0)   execution time : 10.587 s
Press any key to continue.
|
```

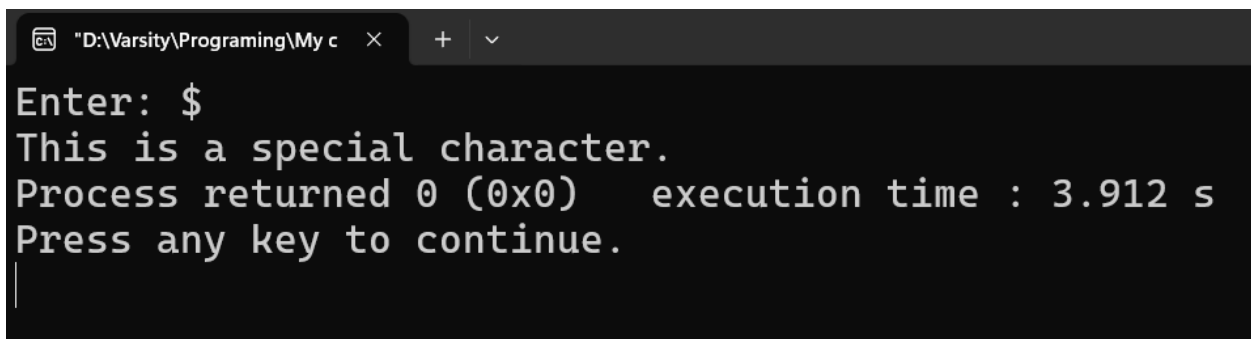
Question 9: Write a C program to input any character and check whether it is alphabet, digit or special character.

```
#include<stdio.h>

int main(){
    char x;
    printf("Enter: ");
    scanf("%c",&x);
    if(x>='A' && x<='Z'){
        printf("This is an uppercase letter.");
    }
    else if(x>='a' && x<='z'){
        printf("This is a lowercase letter.");
    }
}
```

```
else if(x>='0' && x<='9'){  
    printf("This is a digit");  
}  
else{  
    printf("This is a special character.");  
}  
return 0;  
}
```

Output:



```
Enter: $  
This is a special character.  
Process returned 0 (0x0)   execution time : 3.912 s  
Press any key to continue.  
|
```

Question 10: Write a C program to check whether a character is uppercase or lowercase alphabet.

```
#include <stdio.h>
```

```
int main() {
```

```
    char ch;
```

```
printf("Enter a character: ");  
if (scanf(" %c", &ch) != 1) {  
    printf("Error reading input.\n");  
    return 1;  
}  
  
if (ch >= 'A' && ch <= 'Z') {  
    printf("\nResult:\n");  
    printf("The character '%c' is an UPPERCASE alphabet.\n", ch);  
}  
else if (ch >= 'a' && ch <= 'z') {  
    printf("\nResult:\n");  
    printf("The character '%c' is a LOWERCASE alphabet.\n", ch);  
}  
else {  
    printf("\nResult:\n");  
    printf("The character '%c' is not a standard English alphabet (A-Z or a-z).\n", ch);  
}  
  
return 0;  
}
```

Output:

```
"D:\Varsity\Programing\My c  × + v
Enter a character: Q
Result:
The character 'Q' is an UPPERCASE alphabet.
Process returned 0 (0x0)   execution time : 2.821 s
Press any key to continue.
|
```